

An unconventional guide to algorithms for Beginners

If you thought algorithms are only for coding wizards, you're in for some fun.

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Look around you. Anywhere. Chances are, you are in an algorithm right now or are surrounded by hundreds or even thousands of them at any given point of time. In fact, probably you are playing one out even while you read this! Algorithms have existed since the inception of human reasoning or probably even before that. In fact, if you observe carefully,

then you will notice that algorithms are the defining set of instructions in an order of execution that defines how a task can be performed.

We believe that there is an algorithm for every task. Some of it may be well-defined and well-documented, while the others remain to be discovered or defined by some bright young soul. Algorithms are not just limited to mathematics or computers alone. Do you like to cook? Pick up that recipe book, it is a fine example of an entire book with nothing but algorithms. Do you need to take a bath? Apply soap-lather well-rinse-repeat. Got mail? Clear the mailbox-open the envelope-read-reply if necessary-move on to the next one. An algorithm must contain a specific set of instructions in a very well defined manner that will give the same output every single time.

And indeed, such algorithms are the building blocks of computer programming and many mathematical computations. It is only by the use of such algorithms that humankind has been able to make technology and machinery reach the current levels of fault tolerance and error free operations. Algorithms help solve problems ranging across a spectrum that is vast. They help us to solve a Rubik's cube as well as deliver advanced medicines to a critical patient. Algorithms help in enabling a robot to play football, or to use your biometric information like facial recognition, retinal scan or fingerprint based authentication to authorize a transaction. Well documented algorithms existed as far back as in the times of the Vedas in India, ranging on a vast number of topics like mathematics, astronomy, medi-



*Windows on Github

Microsoft is transferring the Windows codebase to a single Git repo hosted on Visual Studio Team Services. This has involved scaling Git to extremely large projects and teams with a project called "Git Virtual File System".
<http://dgit.in/WindowsGit>



*AlphaGo wins again

AlphaGo, the AI created to play the game of Go better than anyone alive, has defeated a team made up of five Go champions in a demonstration match on 26 May, to prove once again that AI has arrived.
<http://dgit.in/AIWinGO>



*Apple course

Apple released a free college-level computer coding course designed to teach students how to create apps, a move that's part of the company's push to highlight its role in the US economy.
<http://dgit.in/AplProg>